

Introduction personnelle

Ingénieure simulation et validation avec une base forte en C++/Python et en développement de workflows scientifiques réutilisables pour des systèmes physiques instrumentés.

Je développe des composants de simulation, des chaînes de correction et des workflows de validation qui rendent les résultats plus exploitables pour les équipes techniques.

Je suis particulièrement pertinente pour des postes où développement logiciel, interprétation des résultats, test/validation et amélioration continue des modèles se croisent.

Selected Fit For This Role

- Developed and adapted Geant4/C++ code for system-response and simulation studies on physical systems, including model logic, geometry, and analysis outputs.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction.
- Built reusable Python/C++ workflows in shared technical environments, with strong attention to robustness and traceability.
- Built validation and monitoring workflows that detect anomalies, trace their causes, and feed corrective actions back to the relevant teams.
- Combined test measurements, simulation, and response maps to model the measurement chain and quantify the system impact of observed discrepancies.
- Wrote structured decision-facing technical results that connect models, methods, discrepancies, and proposed improvements.

simulation-software component development, Python/C, QA, testing and validation, technical reporting, and interaction with product teams.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: ROOT / RooFit | Geant4 | Python | Geant4 simulation | NumPy | C/C++ | Workflow Automation (YAML / Shell / CMake) | pandas | statsmodels | matplotlib / seaborn | etc

Methods: beam-test analysis | correlation-based correction | Detector Simulation | event-level correction | production yield workflows | Maximum Likelihood Estimation | Latent-Parameter Estimation | Probabilistic Modelling | Statistical Inference | Signal Processing | etc

Validation: Calibration & Bias Correction | Model Validation | Simulation-to-Data Alignment | Simulation Calibration | Simulation Correction | Monitoring | Data Validation | Root-Cause Analysis | Bias Correction | Calibration | etc

ML: train-test diagnostics | Domain Adaptation | Feature Validation | scikit-learn | CatBoost / Gradient Boosting | GBDT | CatBoost | Reweighting | Feature Selection | CatBoost training | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Developed CatBoost-based selection and validation pipelines, including train/test diagnostics, scan studies, and reproducible model application.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.

Selected Project

Sensor Simulation & Reconstruction

Signal Processing | Simulation Calibration | Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor...

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation.

Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design

Test-Data Validation & Root-Cause Analysis

Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Pulse-Shape Discrimination for Energy Reconstruction

Pulse Shape Discrimination | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction

Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based...

Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design